WELDING AND MATERIAL JOINING PROCESSES - PART II

Thermit & Modern Welding • Soldering & Brazing • Welding Defects

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1. Thermit Welding

Thermit welding is a fusion process in which heat is obtained from a highly exothermic chemical reaction between aluminium powder and a metal oxide, commonly iron oxide. The reaction produces molten metal and aluminium oxide slag. The molten reaction metal is poured into a refractory mould surrounding the joint and fuses the prepared ends.

A representative reaction is: $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3 + \text{heat}$. The reaction is initiated by a special ignition mixture because ordinary flame ignition is insufficient.

1. Align and gap the parts; prepare and preheat the joint if required.
2. Build a refractory mould around the joint and seal leakage paths.
3. Charge the thermit mixture in a refractory crucible and ignite it.
4. Allow reaction separation of molten metal and slag; tap the molten metal into the mould.
5. Permit solidification, remove the mould, chip/finish excess metal and inspect.

- Advantages: independent of electrical supply, can produce very large local heat input, suitable for heavy sections.
- Disadvantages: slow preparation, difficult control, bulky mould/crucible, unsuitable for thin precision work.
- Applications: rail joints, heavy castings, repair of large steel components, crane rails and similar field work.

2. TIG Welding (GTAW)

TIG welding uses a non-consumable tungsten electrode to establish the arc. The weld pool and electrode are protected by inert shielding gas, usually argon or helium. Filler rod is added separately when required. Because there is no flux, the process produces a clean weld with no slag.

2.1 Equipment and process

- Constant-current AC/DC power source, TIG torch, tungsten electrode and shielding-gas system.
- High-frequency or lift-arc starting system; water cooling for high-current torches.
- Separate filler wire/rod where required; foot/hand current control may be used.
- DC is widely used for steels and many alloys; AC is widely used for aluminium/magnesium because of oxide-cleaning behaviour.

2.2 Advantages

- Excellent weld quality, precise heat control and clean appearance.
- No slag; suitable for thin sheets, root passes and reactive/non-ferrous materials.
- Can weld with or without filler metal and in many positions.

2.3 Disadvantages

- Low deposition rate and slower than MIG/SAW for thick sections.
- Requires high operator skill and clean surfaces; shielding is sensitive to wind.
- Equipment and inert-gas cost are higher than simple SMAW.

2.4 Applications

Aircraft and aerospace components, stainless steel fabrication, aluminium and magnesium alloys, chemical/food equipment, thin-wall tubes, high-quality root welding and precision repair.

3. MIG Welding (GMAW)

MIG welding uses a continuously fed consumable wire electrode and externally supplied shielding gas. The wire acts as both electrode and filler. With inert gases the classical term is MIG; active-gas variants are commonly called MAG, though both belong to GMAW.

3.1 Advantages

- High deposition rate, continuous wire feed and high productivity.
- No slag removal; easily mechanized or robotized.
- Good for thin sheet through medium-thick sections and long welds.

3.2 Disadvantages

- Equipment is more complex and less portable than SMAW.
- Shielding gas is affected by wind; surface contamination can cause porosity.
- Wire-feed and parameter control must be maintained carefully.

3.3 Applications

Automobile bodies, structural fabrication, stainless steel and aluminium fabrication, pressure vessels, pipelines, shipbuilding, robotic production and general manufacturing.

4. Submerged Arc Welding (SAW)

Submerged arc welding uses one or more continuously fed bare wire electrodes. The arc and molten pool are completely covered by a granular fusible flux. The flux melts partly to form slag and shielding, while the unused granular flux can often be recovered.

- Advantages: very high deposition rate, deep penetration, high thermal efficiency, little visible arc radiation and low spatter.

- Disadvantages: generally limited to flat/horizontal positions; equipment is less portable; flux handling is required; unsuitable for very thin work.
- Applications: long seams in pressure vessels, pipes, ship panels, beams, heavy structural fabrication and surfacing.

5. Plasma Arc Welding (PAW)

Plasma arc welding is related to TIG but the arc is constricted through a fine nozzle. This produces a high-energy-density plasma jet. The arc may be transferred to the workpiece or non-transferred between electrode and nozzle.

- Advantages: concentrated heat source, high speed, narrow heat-affected zone, stable arc, keyhole capability on suitable thicknesses.
- Disadvantages: higher equipment cost, precise torch alignment and parameter control required, nozzle/electrode maintenance.
- Applications: stainless steel tubes, aerospace components, precision sheet metal, instrumentation and automated high-quality welding.

6. Electron Beam Welding (EBW)

Electron beam welding focuses high-velocity electrons onto the joint. Their kinetic energy is converted to heat on impact, producing a very deep and narrow fusion zone. Conventional high-quality EBW is performed in a vacuum or partial vacuum to prevent electron scattering.

- Advantages: very high energy density, deep penetration in one pass, narrow weld and heat-affected zone, low distortion, excellent precision.
- Disadvantages: expensive equipment, vacuum-chamber limitations for many systems, strict joint fit-up/alignment, X-ray shielding requirements.
- Applications: aerospace and nuclear parts, turbine components, gears, sensors, high-value dissimilar-metal joints and precision components.

7. Ultrasonic Welding

Ultrasonic welding is a solid-state pressure-welding process. Parts are clamped together under static force while high-frequency mechanical vibrations are applied parallel to the interface. The oscillatory shear breaks surface films and produces intimate metallic contact without bulk melting.

- Advantages: very low heat input, short cycle time, no filler/flux, excellent for thin foils, wires and many dissimilar metals.
- Disadvantages: limited thickness and joint size, specialized tooling, surface geometry and clamping are critical.
- Applications: electrical connections, battery tabs, wire terminals, foils, electronics, plastics and thin non-ferrous assemblies.

8. Laser Beam Welding (LBW)

Laser beam welding focuses coherent light to a small spot, giving very high power density. Depending on power density and travel speed, welding occurs by conduction mode or by keyhole mode with deep penetration.

- Advantages: high speed, low distortion, narrow heat-affected zone, precise non-contact energy delivery, easy automation and access to small areas.
- Disadvantages: high capital cost, precise fit-up and beam alignment, reflective materials can be difficult, eye/skin protection and enclosure are essential.
- Applications: automotive tailored blanks and body components, electronics, medical devices, batteries, aerospace parts, precision tubes and high-speed automated seams.

9. Allied Joining Processes

9.1 Soldering

Soldering joins parts by melting a filler metal whose liquidus temperature is generally below 450°C, while the base metals remain solid. The filler wets the cleaned surfaces and is drawn through a close joint by capillary action. Flux removes oxide films and improves wetting.

9.2 Types of soldering

Type	Heat source / feature	Examples
Iron soldering	Heated soldering iron or electric bit transfers heat locally.	Electrical/electronic work, sheet-metal seams.
Torch soldering	Gas flame heats the joint.	Plumbing and larger fittings.
Dip soldering	Assembly is dipped into molten solder bath.	Mass production of terminals and small components.
Wave soldering	Printed circuit board passes over a controlled wave of molten solder.	PCB manufacture.
Reflow soldering	Solder paste is deposited and then melted in a controlled heating cycle.	Surface-mount electronics.

9.3 Brazing

Brazing is similar to soldering but uses a filler metal with liquidus temperature above 450°C and below the melting point of the base metals. The joint is designed with small clearance so molten filler is drawn into the joint by capillary action. Silver-, copper-, brass- and nickel-base brazing fillers are common, depending on the material and service temperature.

9.4 Difference between soldering and brazing

Point	Soldering	Brazing
Filler-metal temperature	Generally below 450°C.	Above 450°C but below base-metal melting point.
Joint strength	Lower; suited to light-duty/electrical service.	Higher; suitable for stronger mechanical joints.
Heat source	Iron, torch, bath, wave, reflow oven.	Torch, furnace, induction, dip or resistance heating.
Distortion	Very low.	Low compared with fusion welding, but greater than soldering.
Typical uses	Electronics, small sheet work, plumbing.	Carbide tips, heat exchangers, tubes, dissimilar metals, bicycle/refrigeration fittings.

10. Welding Defects, Causes and Remedies

Defect	Typical causes	Remedies / prevention
Porosity / blowholes	Moist electrode/flux, contaminated surface, inadequate shielding, long arc, excessive gas turbulence.	Dry consumables, clean joint, correct gas flow and arc length, avoid drafts.
Slag inclusion	Poor cleaning between passes, low current, incorrect electrode angle, narrow groove.	Remove slag, provide correct groove, increase heat input appropriately, manipulate electrode correctly.
Lack of fusion	Low heat input, fast travel, oxide/scale, wrong electrode angle.	Increase current or reduce speed, clean surfaces, correct technique and joint preparation.
Incomplete penetration	Insufficient root gap, low current,	Modify root preparation/gap, use

Defect	Typical causes	Remedies / prevention
	large root face, wrong electrode size.	suitable current/electrode, back-gouge where necessary.
Undercut	Excessive current, long arc, fast travel, incorrect angle.	Reduce current/arc length, adjust travel speed and angle, use multi-pass technique if required.
Overlap	Too low travel speed, low current, poor electrode manipulation.	Use correct speed/current and electrode angle.
Hot cracking / solidification cracking	High restraint, susceptible composition, poor bead shape, impurities.	Use suitable filler, control joint design/heat input, reduce restraint, maintain proper bead profile.
Cold / hydrogen cracking	Hydrogen, hard microstructure, high restraint, rapid cooling.	Use low-hydrogen consumables, preheat, control heat input, dry electrodes, post-heat where specified.
Distortion / warpage	Unequal heating and contraction, excessive heat input, poor sequence.	Balanced welding, clamps/fixtures, presetting, intermittent/back-step sequence, lower heat input.
Spatter	Excess current/voltage, wrong polarity, unstable arc, poor gas setting.	Correct parameters, polarity, arc length and shielding; maintain equipment.
Burn-through	Excessive heat input, large root gap, slow travel on thin sheet.	Reduce current, increase speed, improve fit-up, use backing or pulse control.

11. Comparison of Major Welding Processes

Process	Electrode / heat source	Shielding	Main strength	Main limitation
SMAW	Flux-coated consumable electrode	Flux gas + slag	Portable and versatile	Low deposition, slag removal
TIG	Non-consumable tungsten	Argon/helium	Highest control/quality	Slow, skill intensive
MIG/GMAW	Continuous consumable wire	External gas	High productivity, automation	Gas sensitive to wind
SAW	Continuous wire under granular flux	Granular flux	Very high deposition	Mostly flat/horizontal
PAW	Constricted tungsten plasma arc	Plasma + shielding gas	Concentrated, stable arc	Cost/complexity
EBW	Focused electron beam	Usually vacuum	Deep narrow weld, low distortion	Expensive vacuum equipment
LBW	Focused laser beam	Optional shielding gas	High speed and precision	High capital cost
Ultrasonic	High-frequency mechanical vibration + pressure	None	No bulk melting, excellent for foils	Limited thickness/size

12. Important Points

- TIG uses a non-consumable tungsten electrode; MIG/GMAW uses a continuously fed consumable wire.
- Submerged arc welding covers the arc with granular flux and is suited to high-deposition long seams.
- Plasma arc is a constricted arc process with higher energy density than ordinary TIG.
- Electron beam and laser beam welding are high-energy-density processes capable of narrow, deep welds.

- Ultrasonic welding is a solid-state process using pressure plus high-frequency vibration.
- Soldering uses filler metal below about 450°C; brazing uses filler above 450°C while the base metal remains solid.
- Porosity, slag inclusion, lack of fusion, incomplete penetration, undercut, cracking and distortion are common welding defects.